

Theory of Computation and Machines Exam

Questions and Answers

Azarbaijan Shahid Madani University - First Semester of
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Question 1

In the theory of computation, what concept does the Church-Turing thesis express?

1. It only applies to Turing machines.
2. **It gives a precise definition of the concept of an algorithm.** (Correct option)
3. It is a mathematical theorem.
4. It has been refuted by Gödel.

Question 2

We call a set of natural numbers recursive if:

1. It is countable.
2. It is infinite.
3. It is a subset of rational numbers.
4. **Its characteristic function is computable.** (Correct option)

Question 3

What does the Universal Turing Machine do?

1. Accepts all languages.
2. Only accepts regular languages.
3. **Simulates any other Turing machine.** (Correct option)
4. Accepts recursive languages.

Question 4

Which of the following statements about languages is incorrect?

1. **The set of recursively enumerable languages is closed under complementation.** (Correct option - incorrect statement)
2. Every recursive language is recursively enumerable.
3. The set of recursive languages is closed under intersection.
4. The halting problem is recursively enumerable.

Question 5

What is a characteristic feature of a k -tape Turing machine?

1. **It can compute faster, but has the same computational power.** (Correct option)
2. It has more computational power than a single-tape machine.
3. All of the above.
4. It cannot be simulated by a single-tape machine.

Question 6

Which of the following statements regarding functions is incorrect?

1. **Every function is TM-computable.** (Correct option - incorrect statement)
2. Every elementary recursive function is TM-computable.
3. None of the above.
4. Every elementary recursive function is total.

Question 7

Which of the following is not closed under complementation?

1. The set of recursive languages.
2. None of the above.
3. **The set of recursively enumerable languages.** (Correct option)
4. The set of regular languages.

Question 8

Which concept is different from the rest?

1. **Recursively enumerable and co-recursively enumerable** (Correct option)
2. Efficiently enumerable
3. Semi-decidable
4. Recursively enumerable

Question 9

What is true about a Turing machine with a two-way infinite tape?

1. **It is equivalent to a Turing machine with a one-way infinite tape.** (Correct option)
2. It only accepts context-free languages.
3. All of the above.
4. It is stronger than a Turing machine with a one-way infinite tape.

Question 10

What is the starting configuration for the string abb ?

1. $q_0 abb$
2. $h\underline{B}abb$
3. None of the above.
4. $q_0\underline{B}abb$ (Correct option)

Question 11

In proving that multiplication is elementary recursive, which of the following was used?

1. **All of the above** (Correct option)
2. The constant function being elementary recursive
3. Induction
4. Composition

Question 12

Which option is not a feature of a two-tape Turing machine?

1. It has a faster computation speed than a single-tape machine.
2. None of the above.
3. **It has greater computational power than a single-tape machine.** (Correct option - not a feature)
4. It can be simulated by a single-tape Turing machine.

Question 13

A function f is called computable if:

1. Its domain is a recursive set.
2. **There exists a Turing machine that computes the output $f(x)$ for every input.** (Correct option)
3. All of the above.
4. It is defined only on natural numbers.

Question 14

If f is uncomputable, then:

1. All of the above.
2. **For every algorithm A , there exists an input n from the domain of the function on which A does not halt.** (Correct option)
3. There is no algorithm that halts for every input n from the domain of the function.
4. There is no algorithm that halts for every n .

Question 15

If language L and its complement are both recursively enumerable, then:

1. It is necessarily context-free.
2. None of the above.
3. It is necessarily regular.
4. **It is necessarily decidable.** (Correct option)

Question 16

Can we construct a Turing machine that decides whether another Turing machine halts on all inputs?

1. Yes, this problem is decidable.
2. Only for Turing machines with limited memory.
3. Only if the input machine has no inputs.
4. **No, this problem is undecidable.** (Correct option)

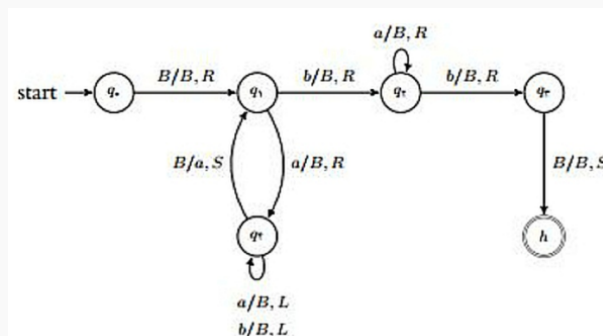
Question 17

Which statement about decidable languages is incorrect?

1. Recursive sets are equivalent to decidable languages.
2. **There exists a Turing machine that halts in finite time for every input.**
(Correct option - incorrect feature)
3. Every decidable language is semi-decidable, but the converse is not true.
4. The complement of a decidable language is decidable.

Question 18

What language does the Turing machine in the image generate?



1. ba^*b (Correct option)
2. ab^*b
3. $(bab)^*$
4. None of the above

Question 19

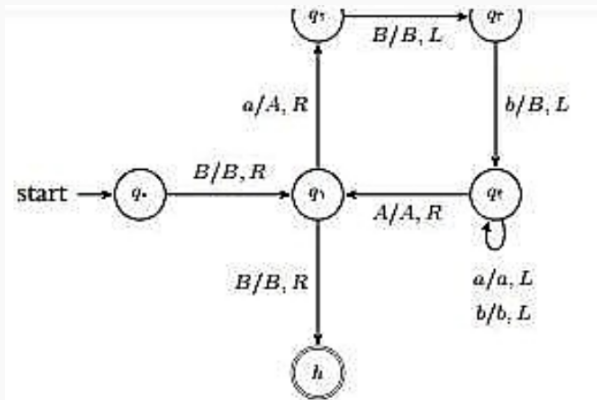
Which statement is incorrect?

1. **Full processing of the string is required for acceptance in a Turing machine.**
(Correct option - incorrect statement)

2. None of the above.
3. Turing machines are both language acceptors and function computers.
4. A TM-computable language is the same as a recursively enumerable language.

Question 20

What language does the automaton in the image accept?



1. Pal
2. None of the above.
3. $a^n b$
4. $a^n b^n$ (Correct option)

Question 21

Which statement about recursively enumerable sets is incorrect?

1. Every recursively enumerable set is accepted by some Turing machine.
2. The halting problem is a recursively enumerable set.
3. **The complement of any recursively enumerable language is recursively enumerable.** (Correct option - incorrect statement)
4. Every recursive set is recursively enumerable.